Classical mechanics as the high-entropy limit of quantum mechanics

Gabriele Carcassi[1], Manuele Landini[2] and Christine A. Aidala[1]

[1]Physics Department
University of Michigan

[2]Institut für Experimental Physik und Zentrum für Quantenphysik,
Universität Innsbruck

Assumptions of Physics

Textbook introduction to quantum mechanics

There were problems with classical mechanics

Blackbody radiation

Photoelectric effect

Stability of atoms

We futzed around with matrices and waves, to find something new that has some correspondence with classical mechanics

$\Rightarrow$ Quantum mechanics!

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Assumptions of Physics

What is the ultimate reason classical mechanics fails, and how is it fixed by quantum mechanics?

Since some of the classical failures are related to thermodynamics, let's look at the relationship between entropy and uncertainty in classical mechanics

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Entropy vs uncertainty

$$S(\rho) = -\int \rho \log h\rho \, dq dp$$

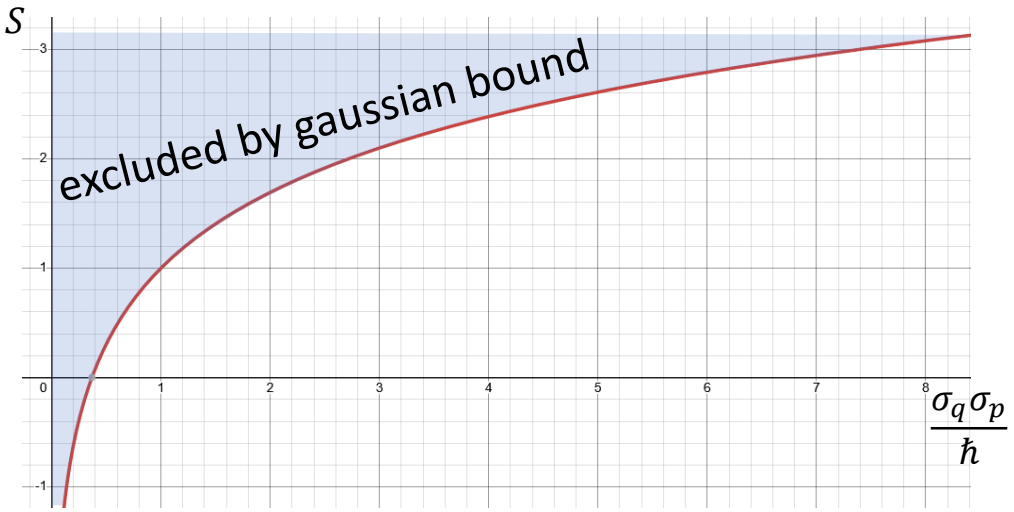
Fixes units

Uniform distribution over volume h has zero entropy

Let's plot one against the other

Gaussian maximizes entropy for a given uncertainty

$$S(\rho) \leq \log 2\pi e \frac{\sigma_q \sigma_p}{h}$$



$$\sigma_q \sigma_p \geq \frac{h}{2\pi e} e^{S(\rho)} = \frac{\hbar}{e} e^{S(\rho)}$$

Entropy puts a lower bound on the uncertainty

Is there a universal lower bound to the entropy?

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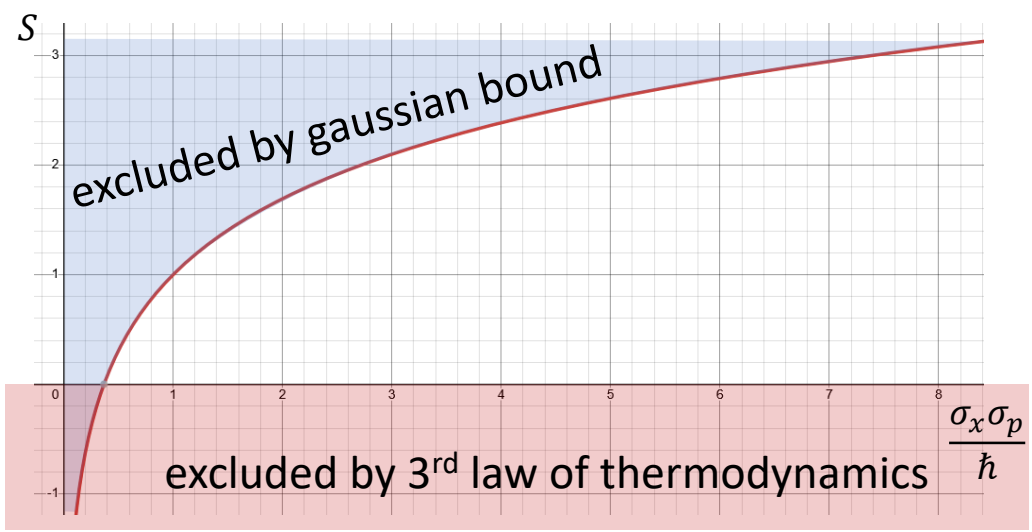
Enter the 3rd law

Every substance has a finite positive entropy, but at the absolute zero of temperature the entropy may become zero, and does so become in the case of perfect crystalline substances.

G. N. Lewis and M. Randall, Thermodynamics and the free energy of chemical substances (McGraw-Hill, 1923)



$$S \geq 0 \quad \Rightarrow \quad \sigma_q \sigma_p \geq \frac{\hbar}{e}$$

Classical uncertainty principle

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Comparing theories

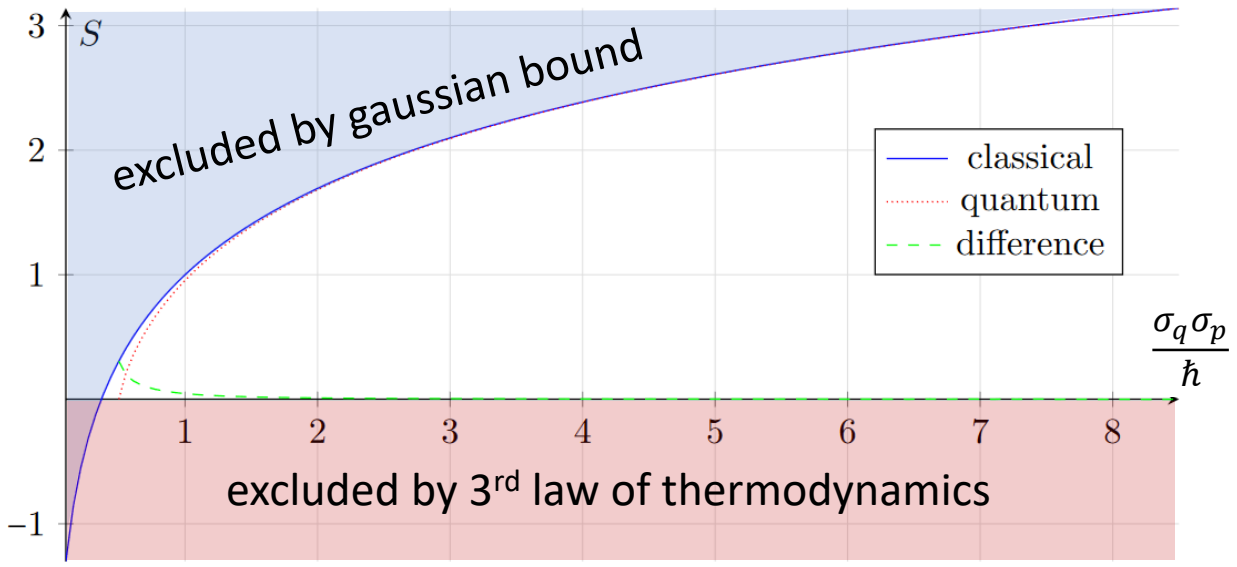
classical

$$\sigma_q \sigma_p \geq \frac{\hbar}{e}$$

2.71828…

quantum

$$\sigma_q \sigma_p \geq \frac{\hbar}{2}$$



Entropy of quantum states is already non-negative

The gaussian bound quickly becomes very similar across theories

$$S_C = \ln e\sigma$$

$$S_Q = \left(\sigma + \tfrac{1}{2}\right)\ln\left(\sigma + \tfrac{1}{2}\right) - \left(\sigma - \tfrac{1}{2}\right)\ln\left(\sigma - \tfrac{1}{2}\right)$$

Quantum mechanics incorporates the third law

Classical mechanics does not

Conjecture: does quantum mechanics recover classical mechanics at high entropy?

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Classical mechanics as high entropy limit?

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greetings

ML Manuele Landini <manulando@gmail.com>
To carcassi@umich.edu

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Wed 7/10/2024 6:15 AM

You replied to this message on 7/10/2024 10:00 AM.

Caro Gabriele,

Mi chiamo Manuele Landini e lavoro a Innsbruck (Austria) come senior scientist in un gruppo di fisica atomica sperimentale. Puoi vedere di cosa ci occupiamo sul nostro sito: https://quantummatter.at.

Ho visto un po' dei tuoi video su youtube. Mi sembra un progetto molto ambizioso, ma promettente. Mi farebbe piacere riuscire a spiegare agli studenti in futuro in termini piu' fisici concetti come le sovrapposizioni o il teorema spin-statistica.

Per la storia della metrica, da quel che ho capito hai bisogno di una metrica che non sia basata sull'entropia, visto che vuoi definire una distanza a entropia costante. Ci sono varie opzioni, ma la trace distance Trace distance - Wikipedia funziona perche' ha una propireta' fondamentale che puoi usare. Chiamala: T(rho,sigma)

Se parti da stati puri, si riduce a (1-<psi|phi>)^(1/2). Quindi per massimizzarla, scegli due stati ortogonali (non importa quali). Il massimo e' T_0=1. Una volta che hai questi stati, che hanno entropia 0, li puoi trasformare in stati con entropia finita (in particolare quelli con massima distanza) tramite una trace preserving map M.

Siccome T si contrae, hai che T(M(rho),M(sigma))<=T(rho,sigma). L'uguale vale se la mappa e' unitaria. Cosi' definisci un serie di step in cui la distanza massima decresce T_n+1<T_n, fino ad arrivare a 0 per stati fully mixed.

arXiv > quant-ph > arXiv:2411.00972

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Quantum Physics

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Classical mechanics as the high-entropy limit of quantum mechanics

Gabriele Carcassi, Manuele Landini, Christine A. Aidala

We show that classical mechanics can be recovered as the high-entropy limit of quantum mechanics. That is, the high entropy masks quantum effects, and mixed states of high enough entropy can be approximated with classical distributions. The mathematical limit $\hbar \to 0$ can be reinterpreted as setting the zero entropy of pure states to $-\infty$, in the same way that non-relativistic mechanics can be recovered mathematically with $c \to \infty$. Physically, these limits are more appropriately defined as $S \gg 0$ and $v \ll c$. Both limits can then be understood as approximations independently of what circumstances allow those approximations to be valid. Consequently, the limit presented is independent of possible underlying mechanisms and of what interpretation is chosen for both quantum states and entropy.

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Assumptions of Physics

Reinterpret old results

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Classical blackbody radiation recovered at small frequencies

$$\frac{2h\nu^3}{c^2}\frac{1}{e^{h\beta\nu}-1} = \frac{2h\nu^3}{c^2}\frac{1}{1+h\beta\nu+O\left(\nu^2\right)-1} \simeq \frac{2\nu^2}{c^2\beta}.$$

At thermal equilibrium, high entropy ⇔ high temperature ⇔ low β

Classical blackbody radiation recovered at small β

$$\frac{2h\nu^3}{c^2}\frac{1}{e^{h\beta\nu}-1} = \frac{2h\nu^3}{c^2}\frac{1}{1+h\beta\nu+O\left(\beta^2\right)-1} \simeq \frac{2\nu^2}{c^2\beta}.$$

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On the Quantum Correction For Thermodynamic Equilibrium

By E. Wigner
Department of Physics, Princeton University
(Received March 14, 1932)

The probability of a configuration is given in classical theory by the Boltzmann formula exp $[-V/kT]$ where V is the potential energy of this configuration. For high temperatures this of course also holds in quantum theory. For lower temperatures, however, a correction term has to be introduced, which can be developed into a power series of $\hbar$. The formula is developed for this correction by means of a probability function and the result discussed.

Wigner function of a thermal state

$$W(x,p) = \int dy e^{i(x+y)p/\hbar} \langle x+y|e^{-\beta \hat{H}}|x-y\rangle e^{-i(x-y)p/\hbar}$$

Classical distribution is the leading term of expansion in $\hbar$

At thermal equilibrium, high entropy ⇔ high temperature ⇔ low β

We recover the same limit and correction with a more meaningful expansion in β

The first-order quantum correction vanishes, showing that the classical approximation is correct to first order in β, which corresponds to the limit of large entropy. We can consider the second-order correction

$$W(x,p) \simeq 1 - \beta\epsilon(x,p) - \beta^2\epsilon^2(x,p) - \beta^2 \int dy \langle x+y|\hat{Q}V(\hat{x}) + V(\hat{x})\hat{Q}|x-y\rangle, \qquad (13)$$

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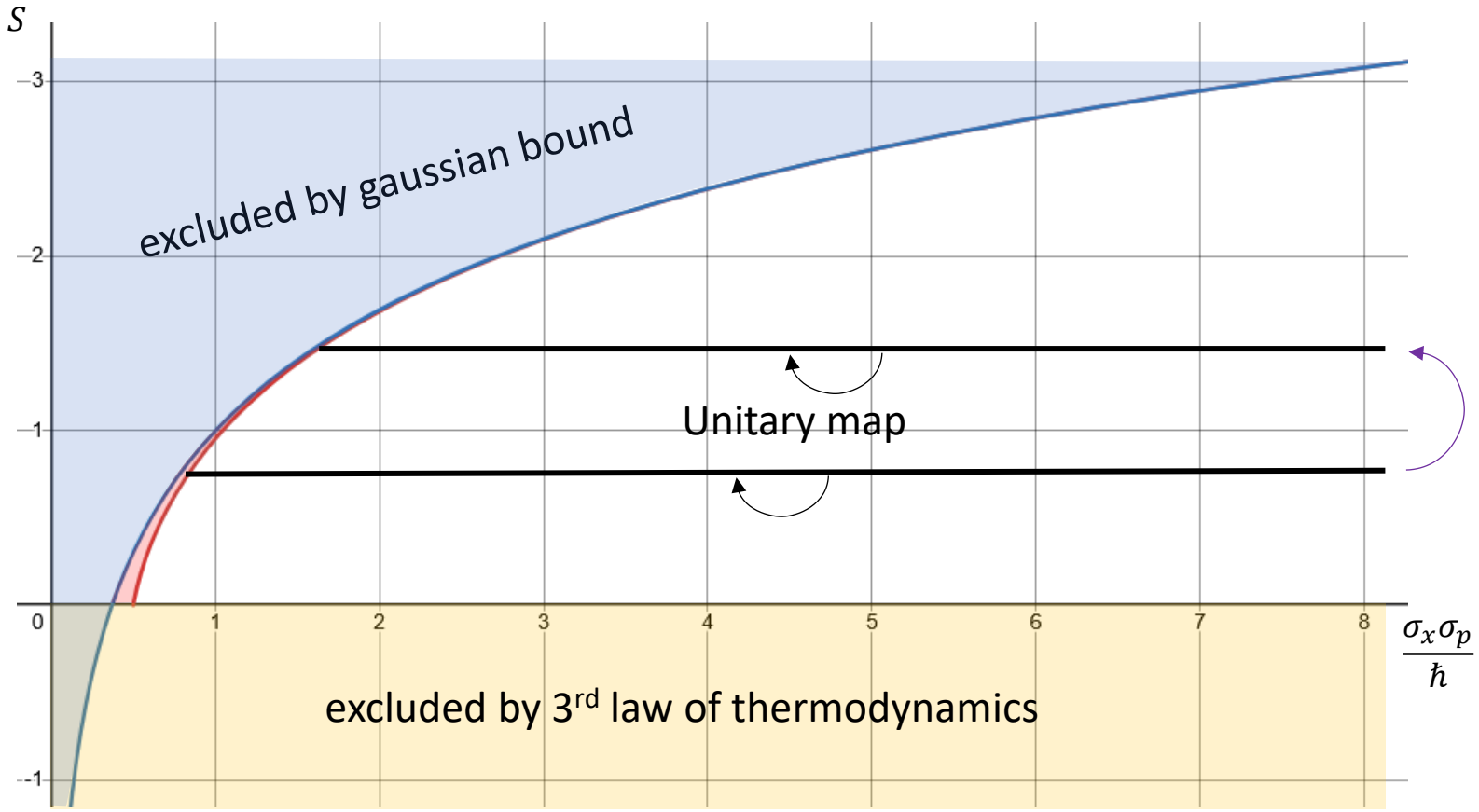
Looking for a map $R(\rho)$ that increases entropy of all mixed states, such that every level set of entropy maps to another level set



⇒ Unitary trans. must be mapped to unitary trans.

$R(\rho)$ Entropy increasing map

classical
quantum

$$[X, P] = \iota\hbar \Rightarrow [T(X), T(P)] = \lambda\iota\hbar$$

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In classical mechanics

$$S\big(R(\rho)\big) = S(\rho) + \log\lambda \Leftrightarrow \{R(x), R(p)\} = \lambda\{x, p\}$$

Jacobian is a constant:
all volumes rescaled
by the same factor

In quantum mechanics

Stretching map $$[R(X), R(P)] = \lambda[X, P]$$

Pure stretching map $$T(X) = \sqrt{\lambda}\, X \qquad T(P) = \sqrt{\lambda}\, P$$

Need to take care of operator ordering!!!

Infinitesimal pure stretching map

$$\frac{dX}{dt} = \frac{\imath}{\hbar}[H, X] + \gamma\left(L^\dagger X L - \frac{1}{2}\{L^\dagger L, X\}\right)$$

Lindblad eq
(open quantum
system)

$$L = a^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(X + \frac{\imath}{m\omega}P\right) \qquad \gamma = \lambda$$

Anti-normal ordering and Husimi Q are preferred

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Effects of stretching map on phase space representations

Husimi Q is simply stretched

Wigner function is stretched and convolved with a gaussian, and coincides with the Husimi Q in the limit

It is interesting to consider what happens to the negative regions of W under the stretching map. We know that W can have negative regions, but their size is limited by the uncertainty principle. In fact, convolving W with a 2D Gaussian with unitary spread, as in the definition of Q, returns a function that is never negative. In the limit $\lambda \gg 1$, the formula for W_λ reduces to

$$W_\lambda(\beta) \to_{\lambda \gg 1} \frac{2}{\pi\lambda^2} \int W_1\left(\frac{\alpha}{\sqrt{\lambda}}\right) e^{-\frac{2}{\lambda}|\alpha-\beta|^2} d^2\alpha = Q_\lambda(\beta). \tag{53}$$

Therefore, while negative regions can be in principle found at any finite λ, W tends to a positive function in the limit. As usual for the W distribution, the phase space size of negative regions is limited to h by the uncertainty principle. The weight of the function in such regions is limited between $\pm 2/h$ for pure states. The effect of the stretching map is to reduce this bound to $\pm 2/(\lambda h)$ for large values of λ. A clear interpretation can be made by working directly on the Fourier transform of W; see Eq. (51). The function $F(W_\lambda)$ is a scaled version of $F(W_1)$ with a Gaussian filter applied to it. Quantum information in W_1 is carried by spectral weights with k-vectors larger than 1. The bandwidth of the Gaussian filter is given by $\lambda/(\lambda-1)$. This cutoff approaches 1 in the limit, filtering away the interference terms.

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Another perspective: move the pure states to minus infinite entropy

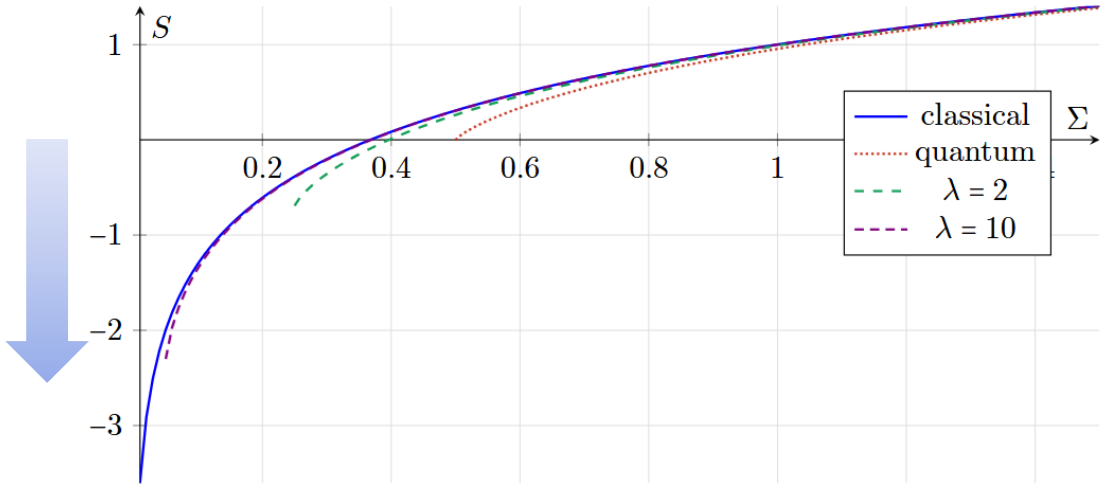
Instead of

$$[X, P] = \imath\hbar \qquad [T(X), T(P)] = \lambda\imath\hbar$$

Redefine original space such that

$$[\hat{X}, \hat{P}] = \frac{\imath\hbar}{\lambda} \qquad [T(\hat{X}), T(\hat{P})] = \imath\hbar$$

$$\lambda \to \infty \;\Rightarrow\; \frac{\hbar}{\lambda} \to 0$$



Mathematically equivalent to lowering the entropy of a pure state to $-\infty$, or $\hbar \to 0$ (group contraction)

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Low speed
"$c \to \infty$"

Speed

High entropy
"$\hbar \to 0$"

Entropy

Classical Mechanics	Relativistic Mechanics
Quantum Mechanics	Quantum Field Theory

No-mechanism limit
(same as non-relativistic limit)

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$$\{A, B\} \longrightarrow \frac{[A, B]}{\imath\hbar}$$

Dirac's correspondence principle: give me a theory with an entropic lower bound that recovers the classical one at high entropy

Only one way to do it

Moyal bracket is the unique one-parameter Lie-algebraic deformation of the Poisson bracket

Quantizing a classical theory means putting a lower bound on the entropy

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In the next 100 years

Textbook introduction to quantum mechanics

Classical statistical mechanics is incompatible with the third law of thermodynamics

Blackbody radiation

Photoelectric effect

Stability of atoms

Let's develop a physical theory that

1. puts a lower bound on the entropy
2. recovers classical mechanics at high entropy

$\Rightarrow$ Quantum mechanics!

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Assumptions of Physics

Assumptions of Physics: find a minimal set of physical assumptions from which the physical laws can be rigorously rederived

- Logic of experimental verifiability ⇒ topologies and σ-algebras
- Real valued quantities recovered from a metrological model
- Variability of elements within an ensemble ⇒ entropy
- Geometric structures ⇔ entropic structures
- Invariance of entropy across observers ⇒ symplectic structure
- Determinism/reversibility ⇔ conservation of entropy ⇔ Hamiltonian evolution
- Geometric and physical interpretation of the action principle
  - Action is the line integral of the vector potential of the flow of states
- Statistical mixing ⇒ all linear structure in physics
- …

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More about our project

- Project website
  - https://assumptionsofphysics.org for papers, presentations, …
  - https://assumptionsofphysics.org/book for our open access book (updated every few years with new results)
- YouTube channels
  - https://www.youtube.com/@gcarcassi
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Assumptions of Physics

Main goal of the project

Identify a handful of physical starting points from which the basic laws can be rigorously derived

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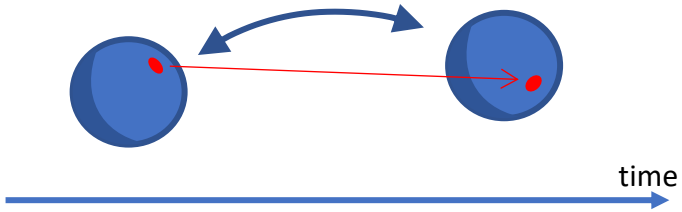
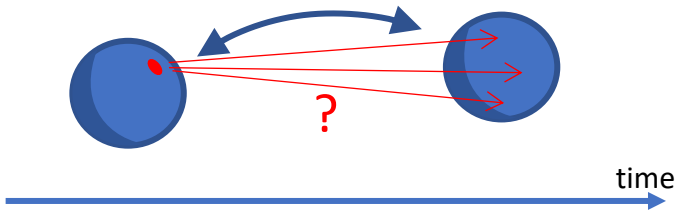
For example:

Infinitesimal reducibility ⇒ Classical state



Irreducibility ⇒ Quantum state



This also requires rederiving all mathematical structures from physical requirements

For example:

Science is evidence based ⇒ scientific theory must be characterized by experimentally verifiable statements ⇒ topology and σ-algebras

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